



Rop Aron Kipyegon

CONTACT INFORMATION	Department of Applied Chemistry University of Debrecen Egyetem Square 1, Hungary	aron.rop@science.unideb.hu www.edu.unideb.hu 06205913567
CITIZENSHIP	Kenyan	
EDUCATION	Ph.D., Chemistry , University of Debrecen, Debrecen, Hungary Dissertation Advisor: Prof Dr. Kéki Sándor Dissertation Title: <i>Intelligent Block Copolymers of Acrylamide Derivatives</i> Msc., Chemistry , University of Debrecen, Debrecen, Hungary Dissertation Advisor: Assoc. Prof Nagy Tibor Dissertation Title: <i>Self-Assembly of Pluronic-Type Amphiphilic Block Copolymers</i> Bsc., Industrial Chemistry , Jomo Kenyatta University of Agriculture and Technology, Kenya Degree conferred with First Class honors.	expected Sep 2028 July 2023 Nov 2018
EMPLOYMENT	PhD Student , University of Debrecen, Debrecen, Hungary	Sep 2024–Present
RESEARCH INTERESTS	My research investigates structure–property relationships in thermo-responsive block copolymers prepared by RAFT polymerization, using a systematic library approach to establish how block length, composition, and dispersity govern LCST cloud point behavior and phase transition kinetics. I am broadly interested in translating these insights toward the rational design of stimuli-responsive materials for smart applications.	
HONORS AND AWARDS	Stipendium Hungaricum , Tempus Public foundation, UD Talent management Program , University of Debrecen,	2021 2022
TEACHING EXPERIENCE	Teching Assistant , University of Debrecen TTKBG0902, Chemical Informatics	Spring 2026
MENTORING/ SUPERVISION	Undergraduate students Ayoade-Alabi Elizabeth Adunola, Chemical Engineering Nassozi Nicole Andrea Crystal, Chemical Engineering	2025 2025

PROFESSIONAL
ACTIVITIES,
OUTREACH, AND
SERVICE**Hungarian Chemical Society, Full member****2025–Present**

COMPUTER SKILLS

Expert in MATLAB, Python, Bash, Microsoft Office, Mnova, Topspin, ChemDraw
Markup languages: L^AT_EX, HTML, CSS, Markdown.PUBLICATION
SUMMARY**Most important Publications —**

1. Gergő Róth, Ákos Kuki, **Aron Kipyegon Rop**, Lajos Nagy, Zuura Kyzy Kaldybek, Máté Benedek, Miklós Zsuga, Tibor Nagy, Sándor Kéki (2025). *Rapid Copolymer Analysis of Unresolved Mass Spectra by Artificial Intelligence*, Analytical Chemistry **2025 97 (36)**, 19801-19808
4. Róth, G. , Kuki, Á. , Nagy, T. , **Rop, AK** , Zsuga, M. , Nagy, L. , Kaldybek, KZ , Benedek, M. , Kéki, S. : *Application of algorithms and neural networks to evaluate block copolymer mass spectra* In: XXXI. International Conference on Chemistry = 31st International Conference on Chemistry, Transylvanian Hungarian Technical Scientific Society, Cluj-Napoca, 48, **2025**, (ISSN 2734-7109)
3. **Rop, AK** , Róth, G. , Pardi-Tóth, VC , Zsuga, M. , Nagy, T. , Kéki, S. : *Rapid Synthesis of Thermoresponsive PNIPAA-b-PNIPAA Block Copolymers*. In: Young Researchers' International Conference on Chemistry and Chemical Engineering (YRICCCE V), Babeş-Bolyai University, Faculty of Chemistry and Chemical Engineering, Cluj., 50, **2025**.
2. **Rop, AK** , Nagy, T. , Róth, G. , Kuki, Á. , Ayoade-Alabi, EA , Nassozi, NAC , Zsuga, M. , Kéki, S. : *Design and Synthesis of Thermoresponsive Copolymers*. In: XXXI. Nemzetközi Kegyésszkonferencia = 31st International Conference on Chemistry, Transylvanian Hungarian Technical Scientific Society, Cluj-Napoca, 47, **2025**, (ISSN 2734-7109)
1. **Rop, AK** , Nagy, T. , Kuki, Á. , Kéki, S. : *Self-Assembly of Pluronic-Type Amphiphilic Block Copolymers*. In: 4th Young Researchers' International Conference on Chemistry and Chemical Engineering YRICCCE IV - **2023**: Program and Book of Abstracts, Hungarian Chemical Society, Debrecen., 68, 2023. ISBN: 9786156018168

REFEREES

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